AMITY SCHOOL OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF AEROSPACE ENGINEERING
AMITY UNIVERSITY MUMBAI

AY 2023-24

SUMMER INTERNSHIP II REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS OF THE DEGREE OF

BACHELOR OF TECHNOLOGY IN AEROSPACE ENGINEERING

CONFERRED BY

AMITY UNIVERSITY MUMBAI

SUBMITTED BY

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SUBMITTED TO

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AMITY SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF AEROSPACE ENGINEERING

This is to certify that **Mr.DARSH YOGESH BUTALA** Enrollment No. **A70405520001** of B.Tech in Aerospace Engineering **Semester-7** has satisfactorily completed the Summer Internship- II prescribed by Amity School of Engineering & Technology, Amity University Mumbai during academic year 2022-23.

Stamp/Seal of the Institute

Name and Signature of Examiner

ACKNOWLEDGEMENT

I convey my sincere gratitude to my Academic Supervisor **Dr. Tushar Chourushi**, Amity University Mumbai. Without his kind direction through the topic and guidance, this study would not have been a success. In every phase of the study, his supervision and guidance shaped this report to be completed perfectly. This experience has been invaluable in enhancing my understanding of Aerospace field and has contributed significantly to my professional development.

Name of the Student: **Darsh Butala**

Enrollment Number: **A70405520001**

CERTIFICATE

OF COMPLETION

This certificate is proudly presented to:

Darsh Butala

*For successfully completing the **4 weeks RC Aircraft Design Training** from 13th July to 13th August 2023.*

Enrollment Number: 2218367792

DANISH SHAIKH
CO-FOUNDER & CEO

FEYNMAN AEROSPACE
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ZAID KARJIKAR
CO-FOUNDER & CTO

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Date: 30th September 2023

ID: FA-AD/0439

CERTIFICATE OF INTERNSHIP

To Whom it May Concern

This is to certify that Mr. Darsh Butala has completed the internship as a 'Aircraft Design Intern' at Feynman Aerospace, from 14th August to 13th September 2023.

During this internship we found him to be active and competent in executing all assigned tasks and also a keen interest and enthusiasm to learn new skills is appreciated.

We wish him success for all the future endeavors.

Danish Shaikh
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ABSTRACT

The objective of this project was to conceptualize a micro-class RC aerial vehicle with a fixed wing configuration. While this aircraft has not been physically constructed, the report encompasses comprehensive details of the manufacturing procedures that could be employed for its fabrication, in addition to an in-depth examination of its overall design, performance, and analysis.

The primary aim of the design was to create an unmanned aerial vehicle capable of carrying a feasible payload. Simultaneously, it aimed to achieve commendable performance characteristics. The modeling phase was executed using SolidWorks, while the analytical aspects were thoroughly assessed employing XFLR5.

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Chapter 1: Introduction

The primary focus of my internship report centered around delving into the intricacies of Aircraft Design. This invaluable learning opportunity was meticulously designed to unravel the diverse facets of aircraft design, with a particular emphasis on airfoil analysis, and comprehensive wing analysis. The overarching aim of the project was to conceptualize, analyze, and construct a Computer-Aided Design (CAD) model of a Micro Aerial Vehicle (MAV).

To fulfill this ambitious objective, an exhaustive research initiative was undertaken. The research aimed at identifying the pivotal parameters essential for crafting a highly efficient and functional aircraft. Central to this exploration was a keen understanding of the significance of airfoil shape, the intricacies of wing design, the imperative of maximizing aspect ratio, the benefits derived from minimizing wing loading, and the strategic optimization of the overall weight of the MAV.

This multifaceted project demanded an in-depth comprehension of airfoil dynamics, considering how the shape influences lift and drag characteristics crucial for the performance of the MAV. Furthermore, an intricate analysis of various wing designs was conducted, exploring the nuances of aerodynamics to ascertain the most optimal configuration.

The emphasis on maximizing aspect ratio in the design process was pivotal, as it directly impacted the MAV's lift-to-drag ratio. This optimization played a pivotal role in enhancing the overall efficiency and performance of the aircraft. Simultaneously, the meticulous endeavor to minimize wing loading contributed to achieving superior aerodynamic performance, ensuring the MAV could operate effectively across a range of conditions.

The comprehensive approach to the project extended to a holistic optimization of the overall weight of the MAV. Striking the right balance between structural integrity, material selection, and weight distribution became a critical aspect of the design process. This overarching optimization aimed to enhance the MAV's maneuverability, fuel efficiency, and overall operational capabilities.

In summary, the internship provided an immersive learning experience, offering insights into the intricate world of aircraft design. The journey encompassed theoretical insights, practical applications of research findings, and the hands-on creation of a CAD model—a testament to the fusion of knowledge and practical skills in the dynamic field of aviation.

1.1: Internship Role/Responsibilities

Task

- Design an aircraft with a wingspan of 150 cm.
- The aircraft weight to be around 1-1.5kg with Speed of 10-15 m/s

During my time at Feynman Aerospace, I was assigned the task of designing an aircraft with a wingspan of 150 cm. The specifications included keeping the weight in the range of 1 to 1.5 kilograms and ensuring a speed of 10 to 15 meters per second.

My main responsibilities involved using XFLR5 software for airfoil analysis. This tool helped in examining lift and drag characteristics, providing data that was then applied to create a CAD model using SolidWorks.

Using XFLR5 allowed for a detailed look at the aerodynamics, helping in making informed decisions about the airfoil shape. This step was essential for achieving a balance between lift and drag to enhance the aircraft's overall efficiency.

Moving on to SolidWorks, the goal was to translate the analyzed airfoil data into a 3D model of the aircraft. This process included detailing the fuselage, wing structures, control surfaces, and other components, ensuring the design was both functional and manufacturable.

In summary, my role at Feynman Aerospace involved a practical blend of software-driven airfoil analysis and CAD modeling. It provided valuable insights into the engineering aspects of aircraft design, highlighting the collaborative relationship between analytical tools and CAD software in achieving design goals.

Chapter 2: Main Wing Analysis

2.1: Main Wing Airfoil Analysis

An airfoil shaped body, moving through a fluid, produces aerodynamic forces, which are lift and drag and it is the airfoil geometry which determines the chord-wise lift distribution of a wing.

Therefore, one of the important design phases of making an aerodynamically efficient aircraft is the selection of the appropriate airfoil.

In the initial stage of analysis, the primary objective was to carefully select a suitable airfoil and subsequently conduct a comprehensive analysis.

My initial inclination was towards utilizing the NACA 2414 airfoil. However, prior to making this selection, I performed a few fundamental calculations, beginning with the determination of the required lift (L) for the aircraft, which is equivalent to its weight, amounting to 1.5 kilograms.

This calculation involved considering the density of air, which stands at 1.293 kilograms per cubic meter, the aircraft's velocity (V) at 10 meters per second, and the surface area of the wing (S), calculated as 1 meter by 1.5 meters, yielding a total of 1.5 square meters.

$$L = \frac{1}{2} \rho V^2 S C_L$$

Where,

L = Lift which is equal to weight = 1.5kg

ρ = Density of air = 1.293 kg/m^3

V = Velocity of Aircraft = 10 m/s

S = Surface area of wing = 1m*1.5m = 1.5m^2

C_L = Coefficient of Lift

The essential parameter of interest in this context was the Coefficient of Lift (CL). Through the application of the lift equation and solving for CL, it was determined that the minimum coefficient of lift required for level flight was 0.015

My initial thought was to go with NACA 2414[1]

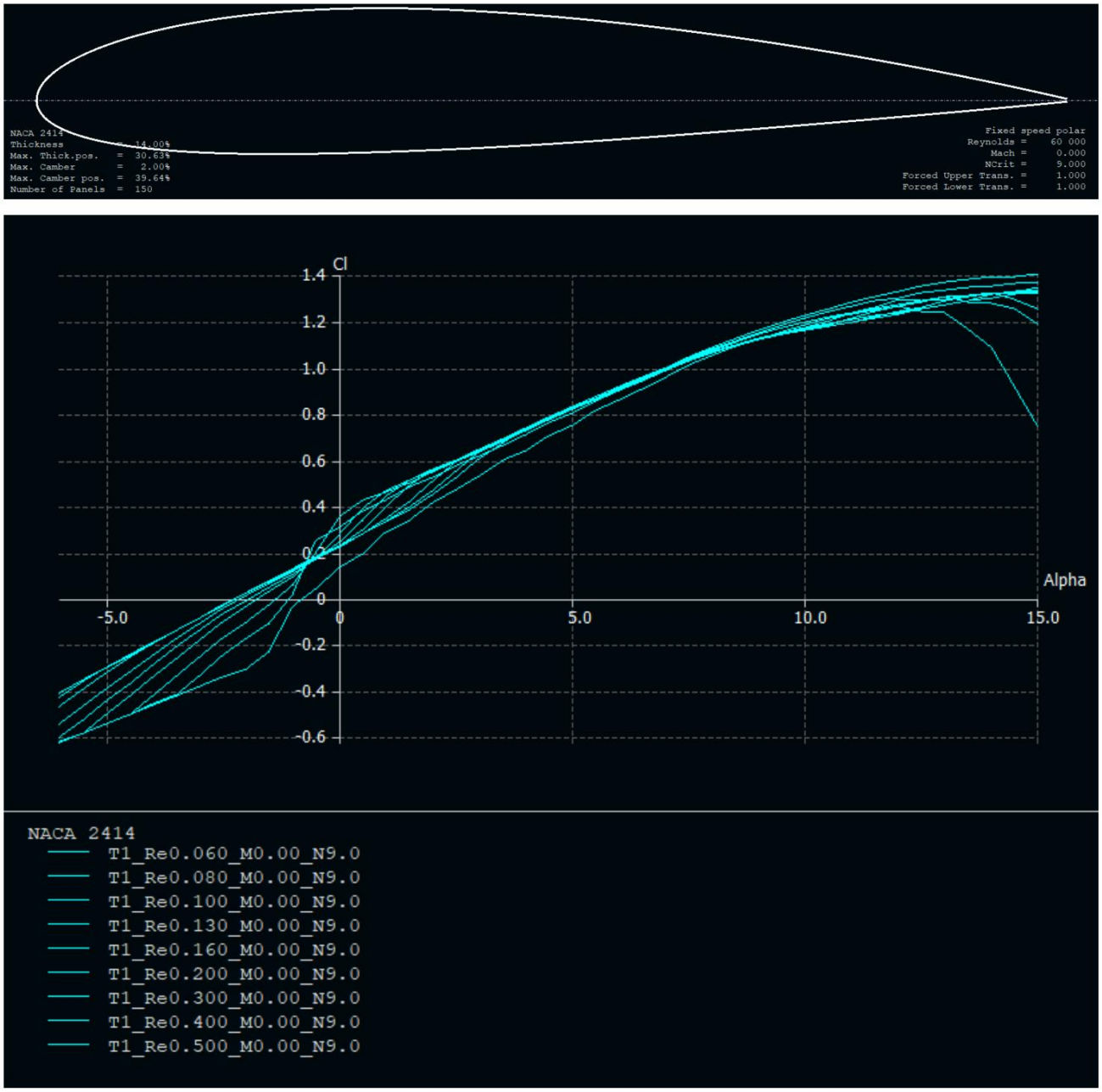


Figure 1: NACA 2414 Analysis

But upon analysis, I observed that the NACA 2414 airfoil displayed a lift coefficient of approximately 0.2 at a zero-degree angle of attack, which appeared to be slightly excessive for the specific requirements of the aircraft. Consequently, I made the decision to transition to a thinner airfoil, specifically the NACA 1412[2].

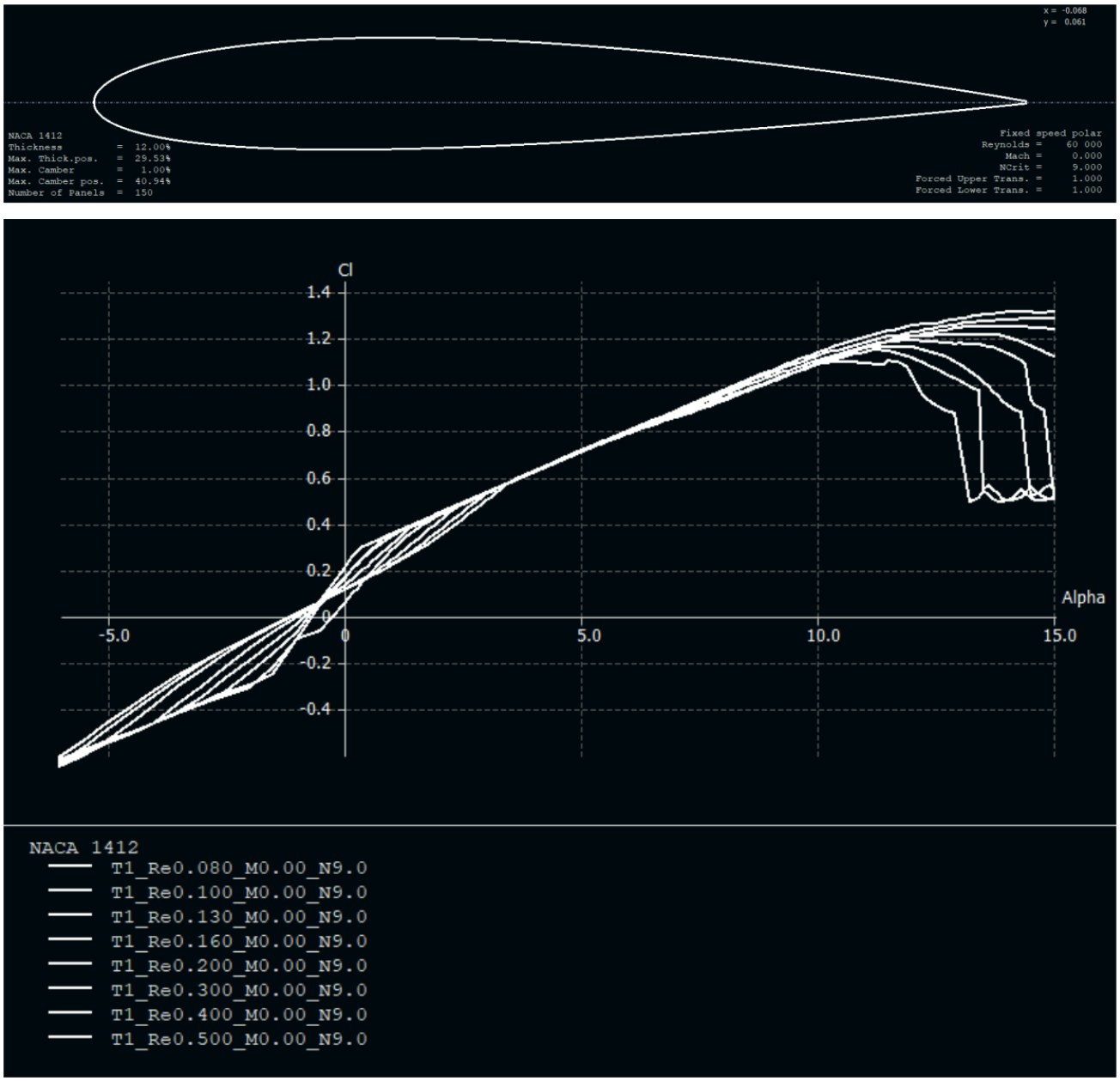


Figure 2: NACA 1412 Analysis

In contrast to the NACA 2414 airfoil, the lift coefficient of the NACA 1412 airfoil was observed to be slightly below 0.2 at a zero-degree angle of attack. This characteristic was more in line with the specific performance requirements identified for the aircraft. The decision to transition to the NACA 1412 airfoil was driven by the aim to better align the aerodynamic properties with the desired performance characteristics, ensuring a more optimal balance between lift and drag for the overall efficiency of the aircraft.

Upon a thorough comparison of the results for both airfoils, it is evident that the lift coefficient of the NACA 1412 is lower when compared to that of the NACA 2414. However, it is crucial to contextualize this observation. The NACA 1412's reduced thickness profile presents noteworthy advantages, notably in terms of cost savings related to materials and a subsequent decrease in wing weight.

The reduction in wing weight holds particular significance as it contributes to enhanced payload capacity for the aircraft. This outcome aligns with the overarching objective of optimizing the aircraft's performance for its intended mission requirements. By prioritizing a thinner airfoil profile, the design not only achieves the desired lift and drag balance but also offers practical advantages in terms of operational efficiency and cost-effectiveness.

In summary, while the NACA 1412 may exhibit a lower lift coefficient, the trade-off in favor of reduced thickness brings about tangible benefits, including cost savings and improved payload capacity, contributing to the overall effectiveness and practicality of the aircraft design.

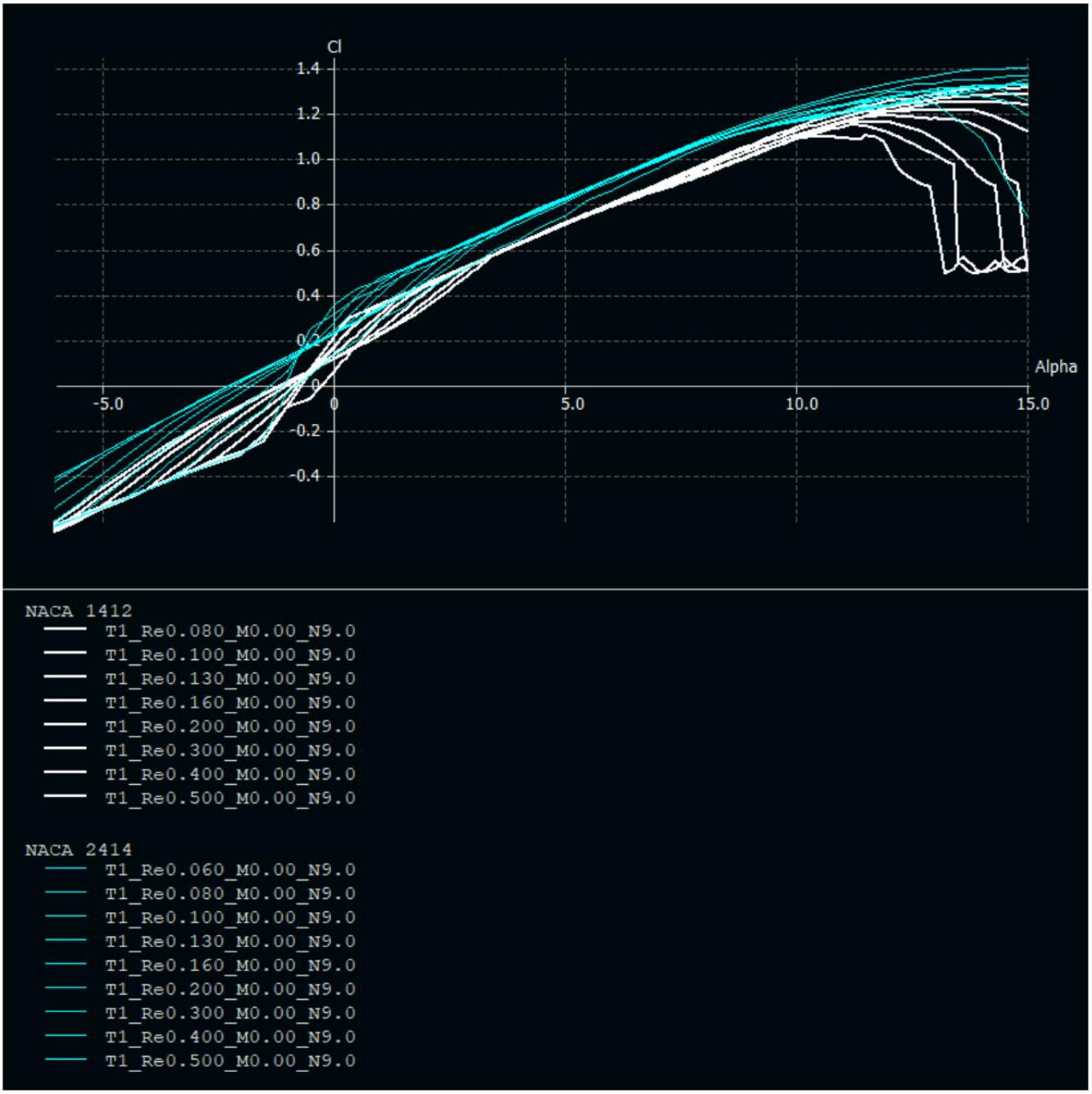


Figure 3: Comparison between NACA 2414 and NACA 1412

Complete Data Analysis of NACA 1412[2]

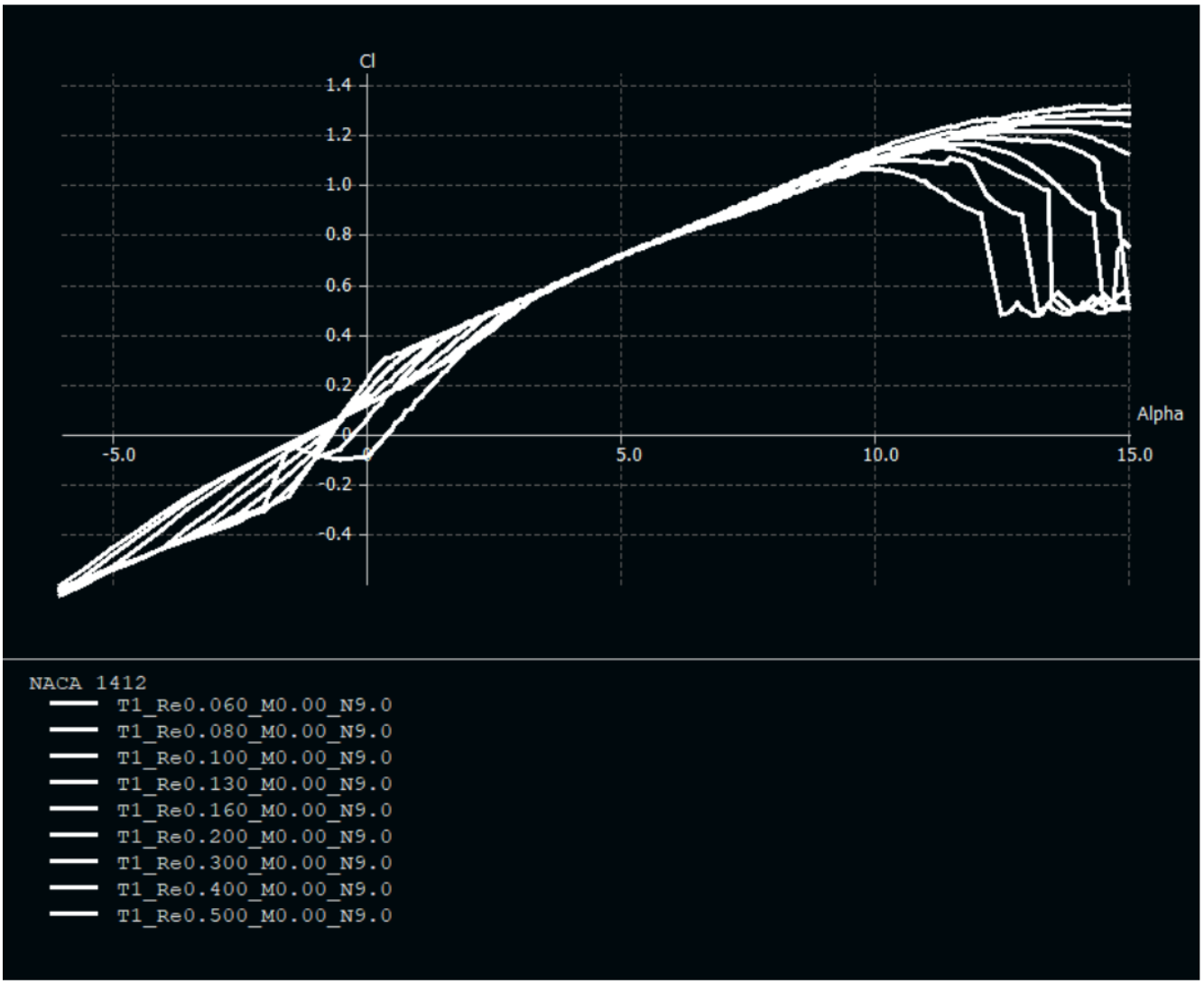


Figure 4: CL vs alpha of NACA 1412

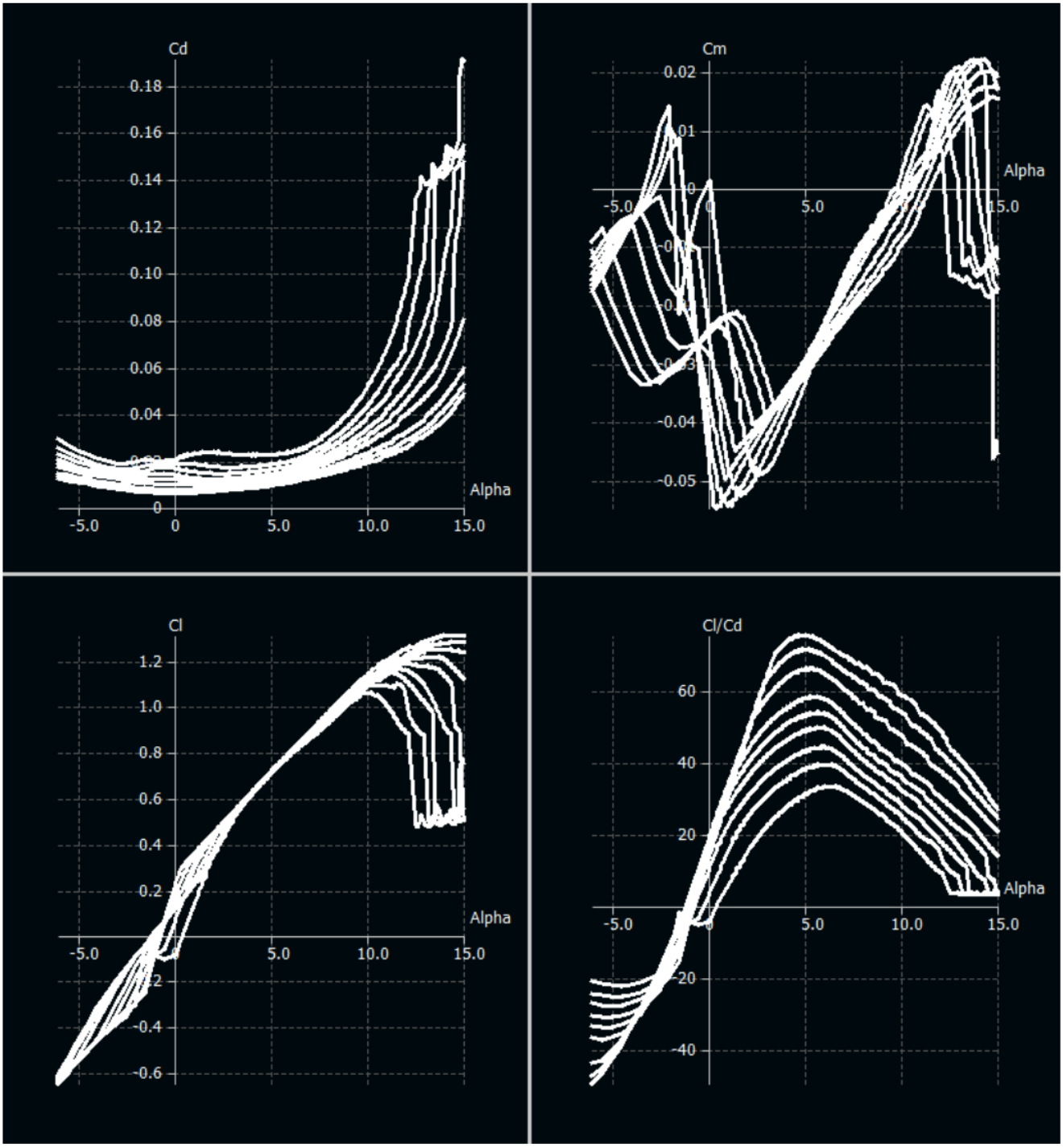


Figure 5: Complete Data of NACA 1412

2.2: Main Wing Airfoil Analysis with Flaps

Subsequently, the next stage in the analysis process entailed a thorough examination of the NACA 1412 airfoil, incorporating the utilization of flaps. These flaps were strategically positioned at an angle of 8 degrees, with the upward orientation denoted by red coloration and the downward orientation represented by green coloration. The placement of these flaps was precisely set at a distance of 70% from the leading edge of the chord.

The introduction of flaps serves as a pivotal aspect of the analysis, as they play a significant role in modifying the aerodynamic characteristics of the airfoil. By strategically adjusting the angle of the flaps, there is an intentional alteration in the lift and drag forces experienced by the airfoil, thereby influencing the overall performance of the aircraft.

The specific placement of the flaps at 70% from the leading edge of the chord is a deliberate choice, as it aims to optimize their impact on the aerodynamics of the airfoil. This positioning allows for a nuanced control of lift and drag, contributing to the fine-tuning of the aircraft's response to different flight conditions.

In summary, the incorporation of flaps into the analysis of the NACA 1412 airfoil marks a critical phase in refining the aerodynamic performance of the aircraft. The strategic placement and adjustment of these flaps at an 8-degree angle provide a means to enhance control and maneuverability, offering valuable insights for optimizing the overall design in accordance with the intended mission requirements.

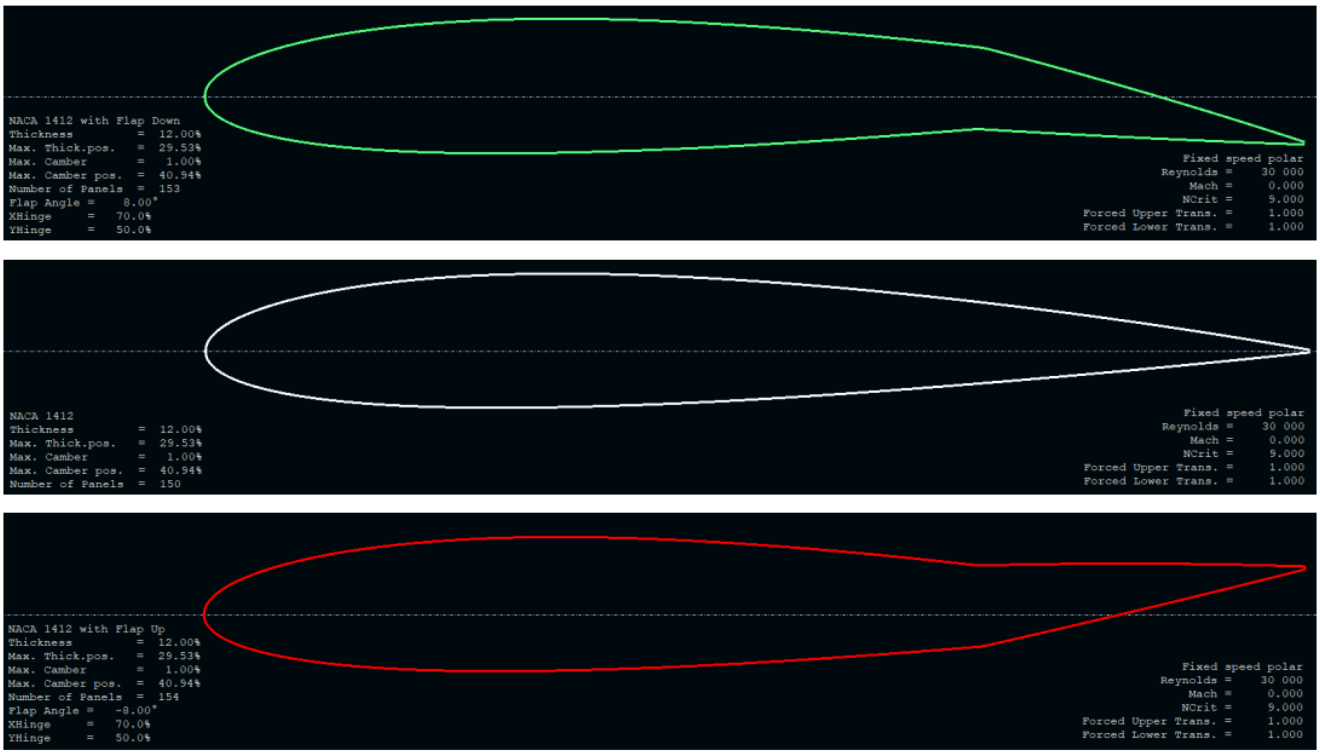


Figure 6: NACA 1412 with Flaps

Figure 7: CL vs alpha of NACA 1412 with flaps

Chapter 3: Tail Wings Analysis

3.1: Horizontal Stabilizer & Vertical Stabilizer Rudder

The selection process for both the Horizontal Stabilizer and Vertical Stabilizer led to the choice of the NACA 0009[3] airfoil. This decision was primarily influenced by the airfoil's symmetrical profile, which guarantees the generation of no lift at a zero-degree angle of attack. Furthermore, the airfoil's thin profile plays a key role in significantly minimizing drag.

Opting for the NACA 0009 airfoil for the stabilizers aligns with the design objective of maintaining stability without introducing unwanted lift forces. The symmetrical nature of the airfoil ensures that, at a zero-degree angle of attack, there is no net upward or downward force generated, contributing to the stability of the aircraft.

Additionally, the thin profile of the NACA 0009 airfoil is advantageous in terms of drag reduction. A thinner airfoil typically experiences less drag, contributing to overall aerodynamic efficiency. This characteristic is particularly valuable for the stabilizers, as it helps optimize their performance without introducing unnecessary resistance during flight.

In summary, the deliberate choice of the NACA 0009 airfoil for both the Horizontal Stabilizer and Vertical Stabilizer underscores a strategic approach to stability and aerodynamic efficiency. The symmetrical profile ensures stability by minimizing lift at zero-degree angle of attack, while the thin profile contributes to drag reduction, enhancing the overall performance of the stabilizing components in the aircraft design.

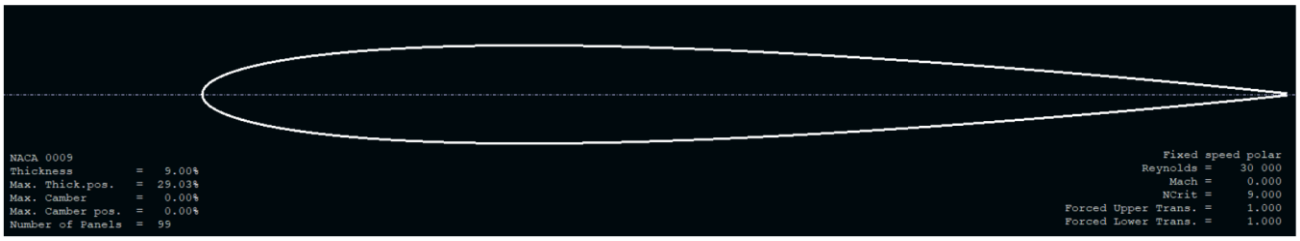


Figure 8: NACA 0009

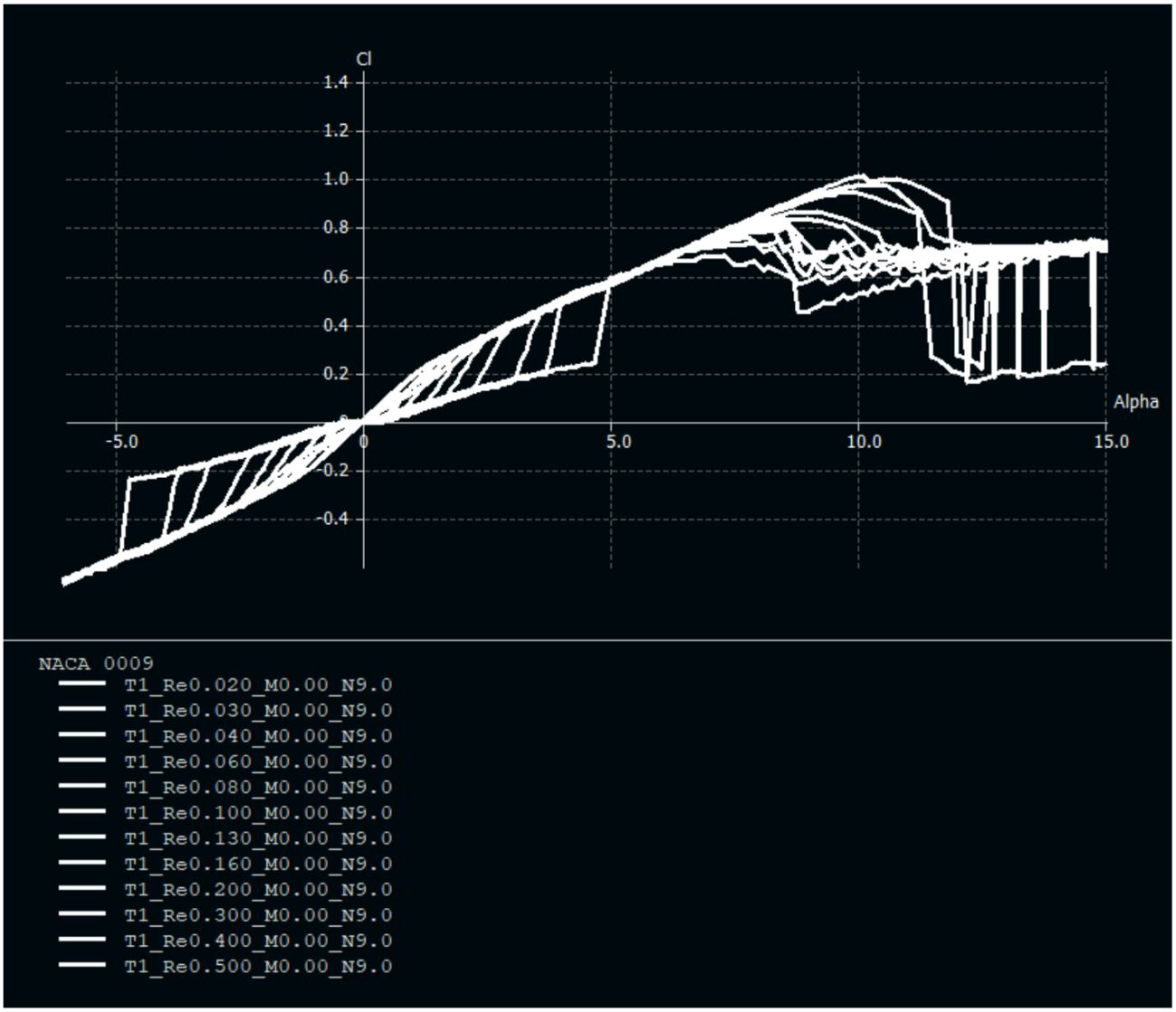


Figure 9: CL vs alpha of NACA 0009

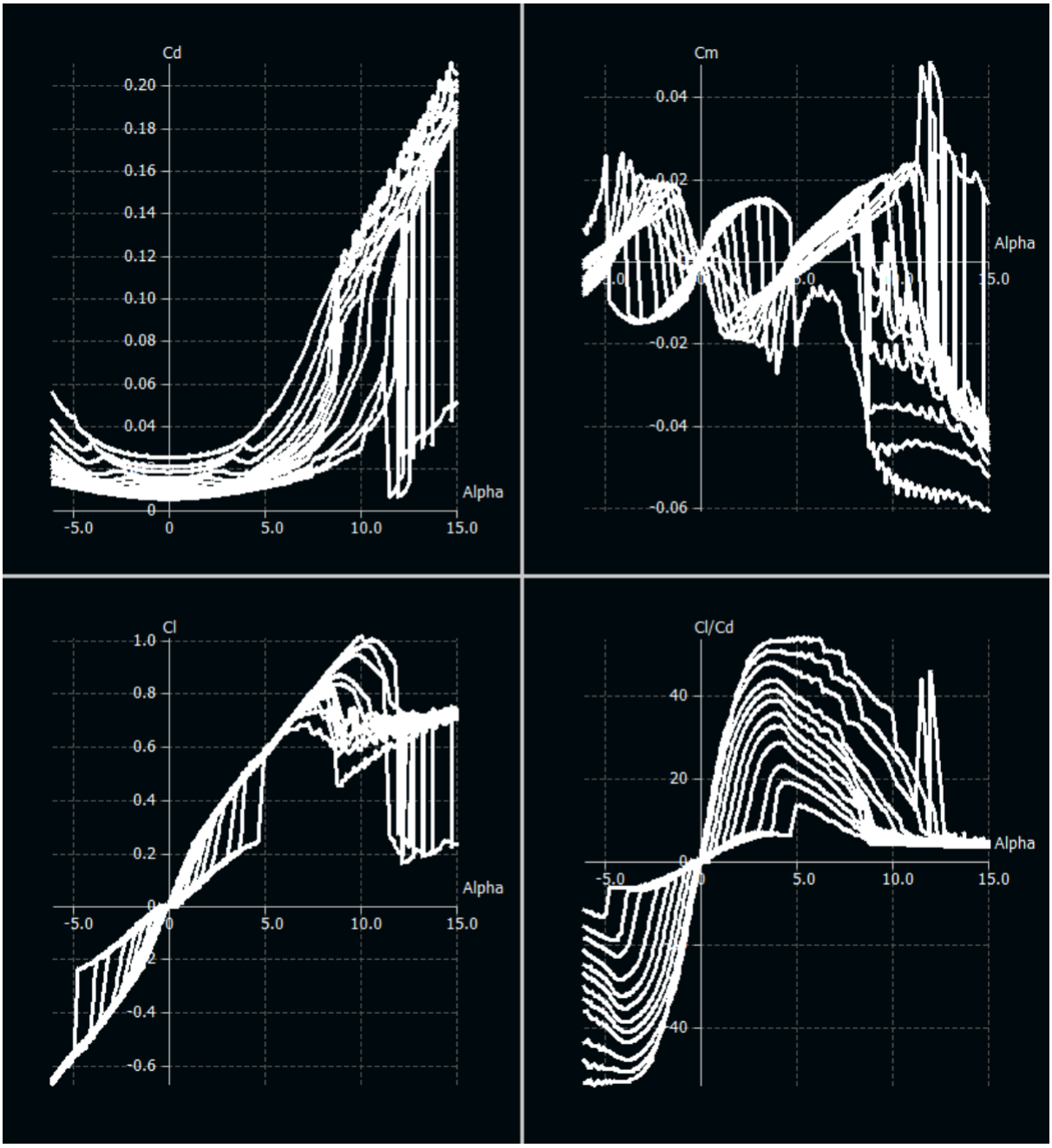


Figure 10: Complete data of NACA 0009

3.2: Horizontal Stabilizer with Elevator & Vertical Stabilizer with Rudder

The Rudder and Elevator components were configured with a deflection of 20 degrees applied to the primary airfoil.

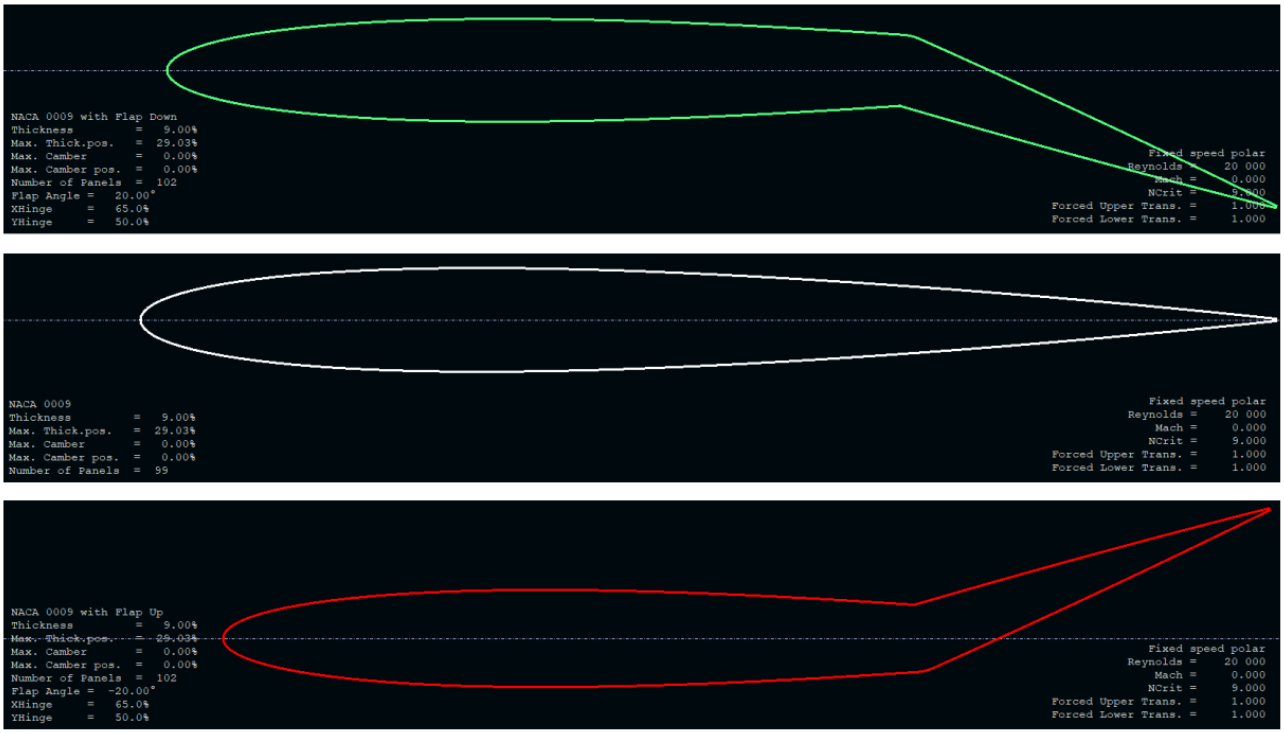


Figure 11: NACA 0009 with flaps

Figure 12: CL vs alpha of NACA 009 with flaps

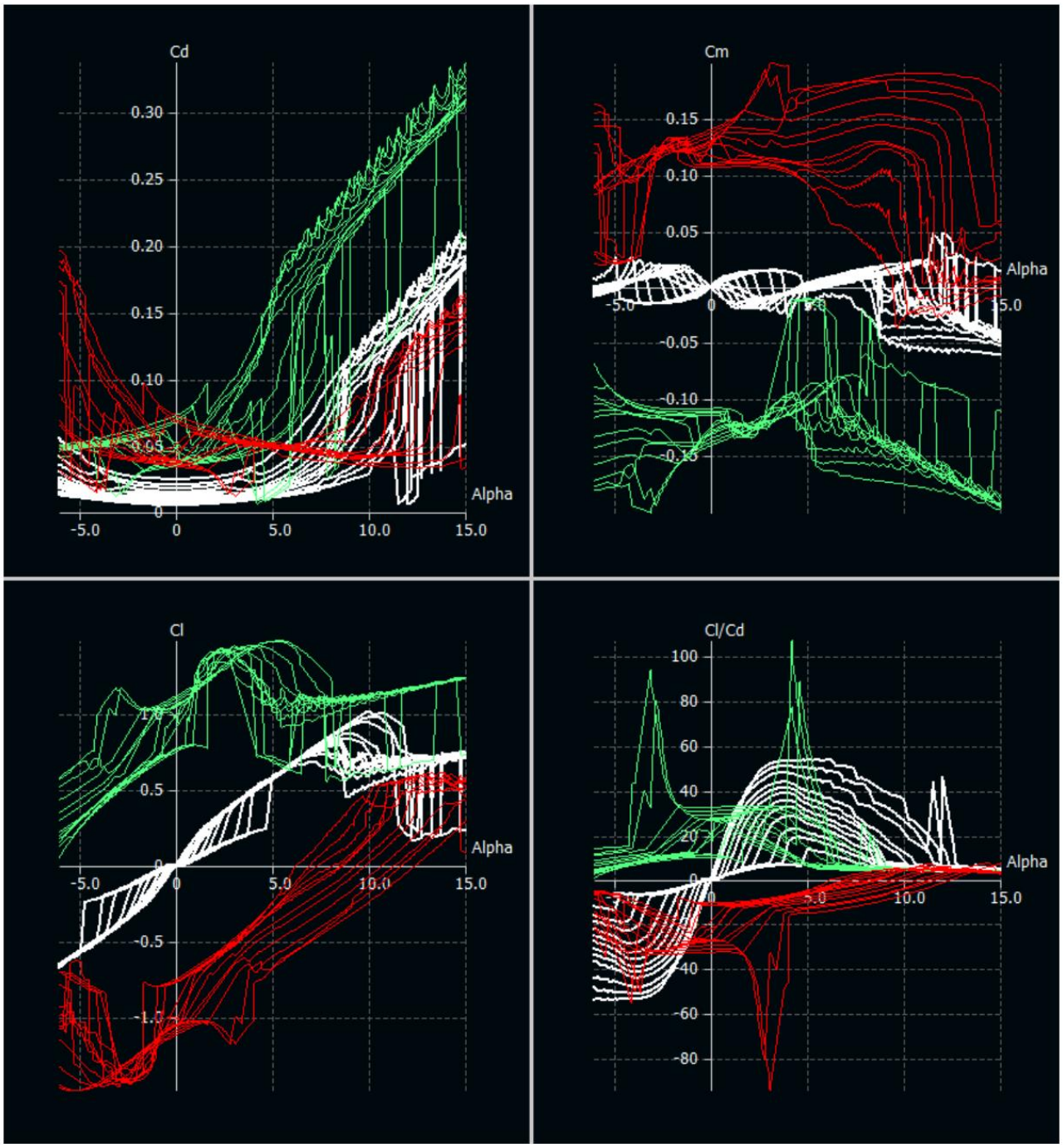


Figure 13: Complete data of NACA 0009 with flaps

Chapter 4: Complete Wing Analysis (Without Fuselage)

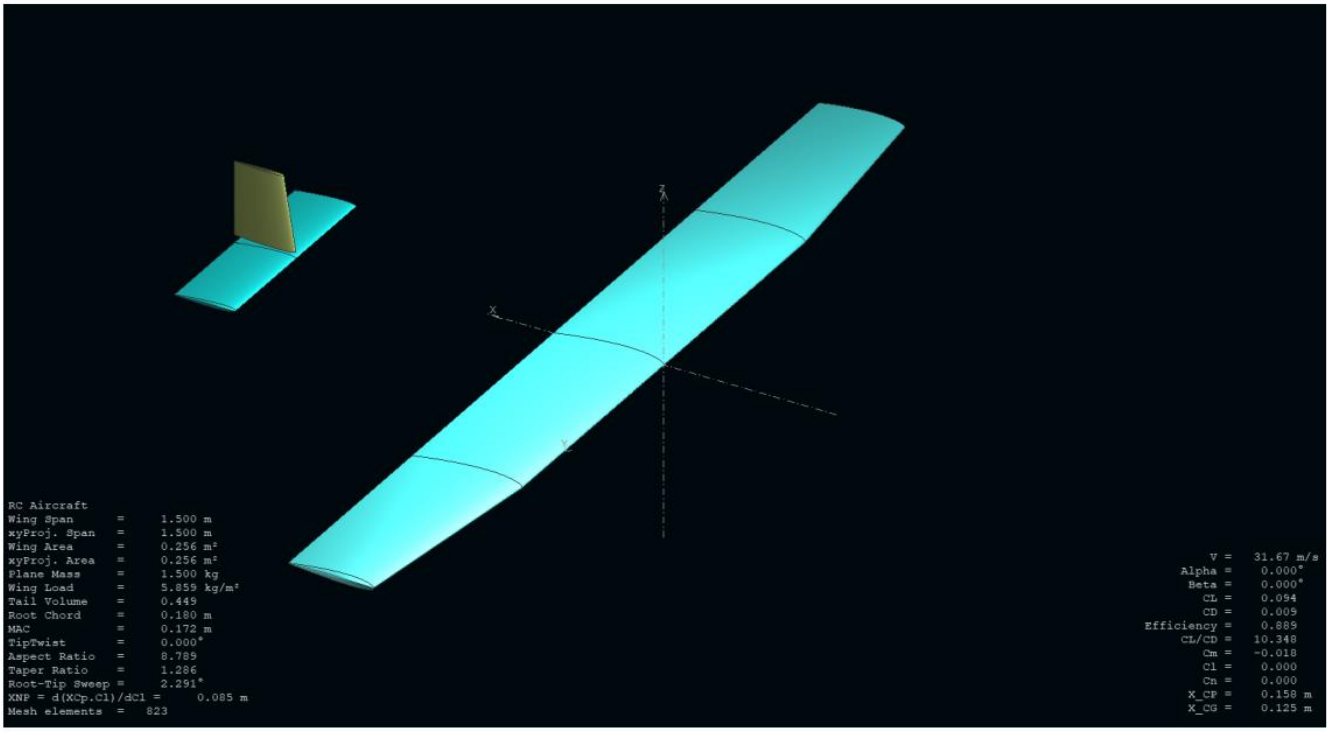
Following the completion of individual airfoil analyses, a thorough and comprehensive wing analysis became imperative, encompassing both the Main wing and Tail wing at their designated positions. The design of these wings incorporated appropriate taper, ensuring a wingspan of 1.5 meters. Additionally, the wings were equipped with the necessary masses to simulate realistic conditions. Throughout this comprehensive analysis, key aerodynamic parameters such as the Coefficient of Lift (CL), Coefficient of Drag (CD), Coefficient of Moment (CM), and the CL/CD ratio were diligently obtained. These parameters were evaluated across varying angles of attack.

This meticulous analysis provided invaluable insights into the aerodynamic performance of the entire wing configuration. The Coefficient of Lift (CL) offered an understanding of the lift generated by the wings, crucial for sustaining flight. Simultaneously, the Coefficient of Drag (CD) illuminated the drag forces acting on the wings, influencing fuel efficiency and overall performance. The Coefficient of Moment (CM) provided insights into the rotational forces around the aircraft's center of gravity.

Evaluating these parameters in relation to varying angles of attack allowed for a comprehensive understanding of how the wing configuration responded to different flight conditions. The CL/CD ratio, in particular, offered a key metric for assessing the trade-off between lift and drag, guiding the optimization process.

This rigorous analysis played a pivotal role in refining the design and optimizing the aircraft's overall efficiency and stability. By gaining a nuanced understanding of how the wings interacted with the air at different angles of attack, the design could be adjusted to enhance performance while ensuring stability under diverse operating conditions. The data obtained from this comprehensive wing analysis served as a crucial foundation for informed decision-making in the ongoing refinement of the aircraft design, contributing to its overall effectiveness and aerodynamic excellence.

Wing Analysis with appropriate Mass added.



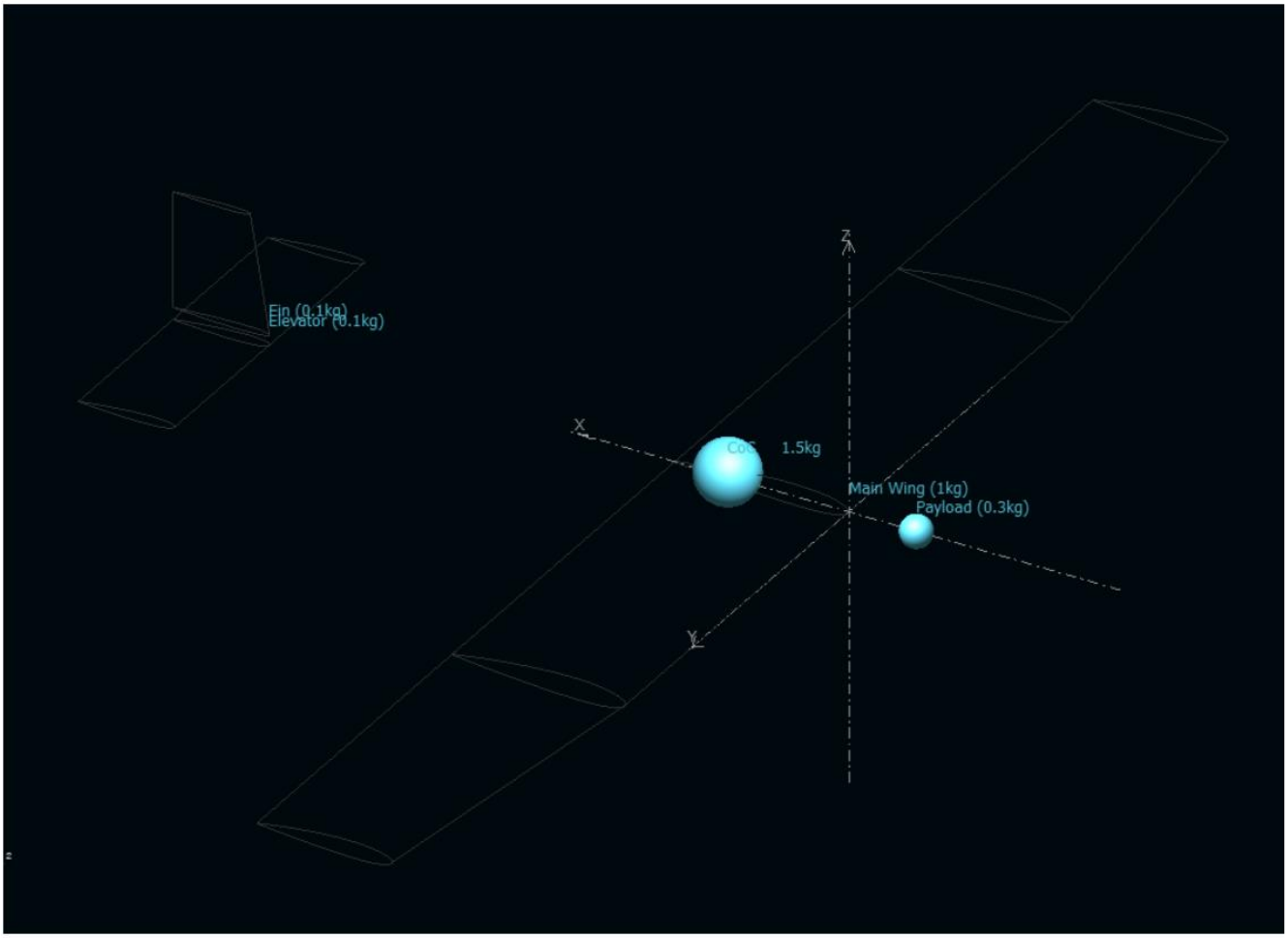


Figure 14: Complete wing analysis

Lift and Induced Drag

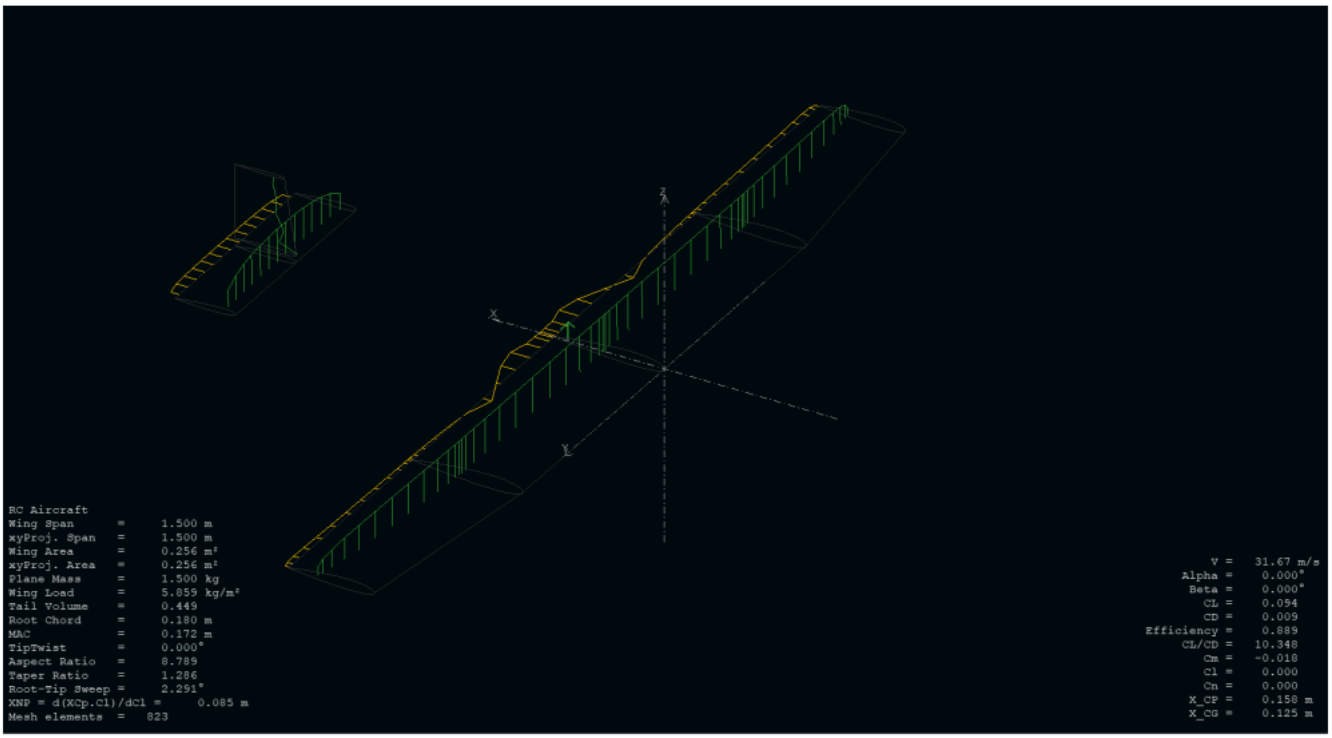


Figure 15: Lift induced Drag

Viscous Drag

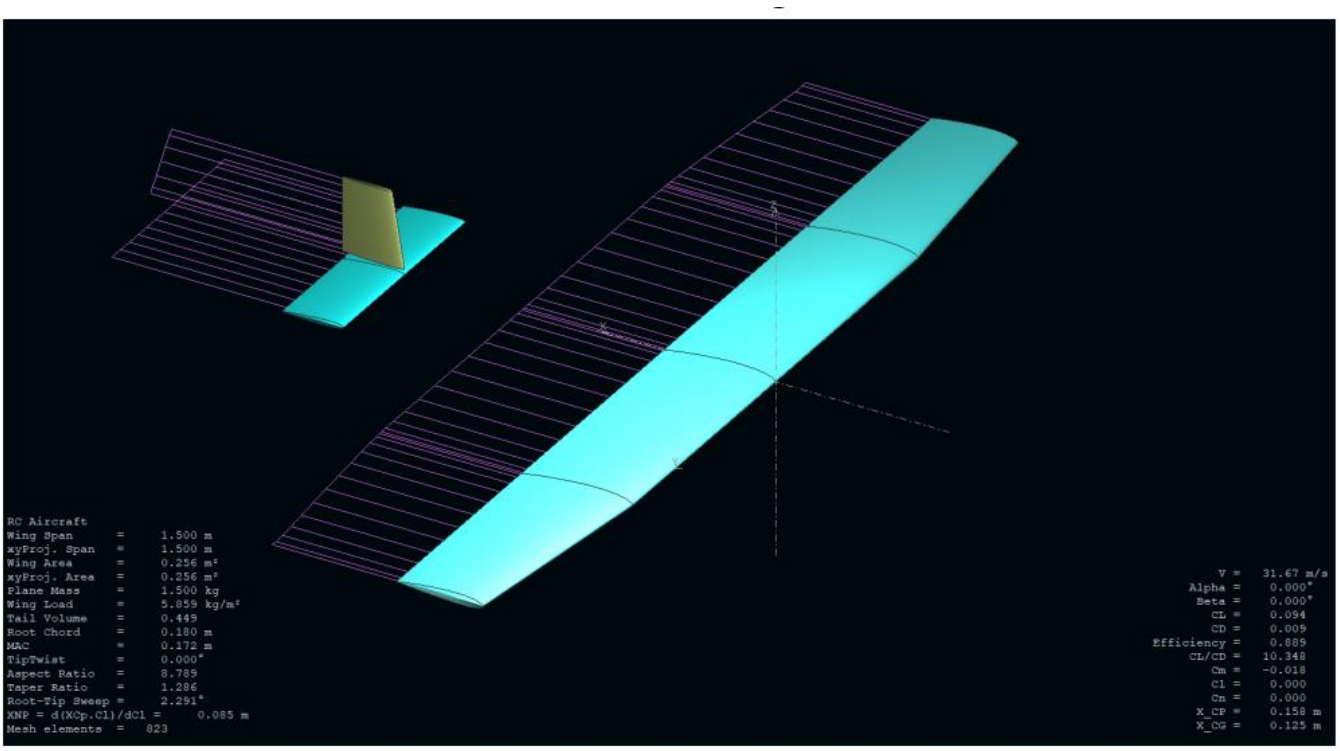


Figure 16: Viscous Drag

Streamlines of Airflow

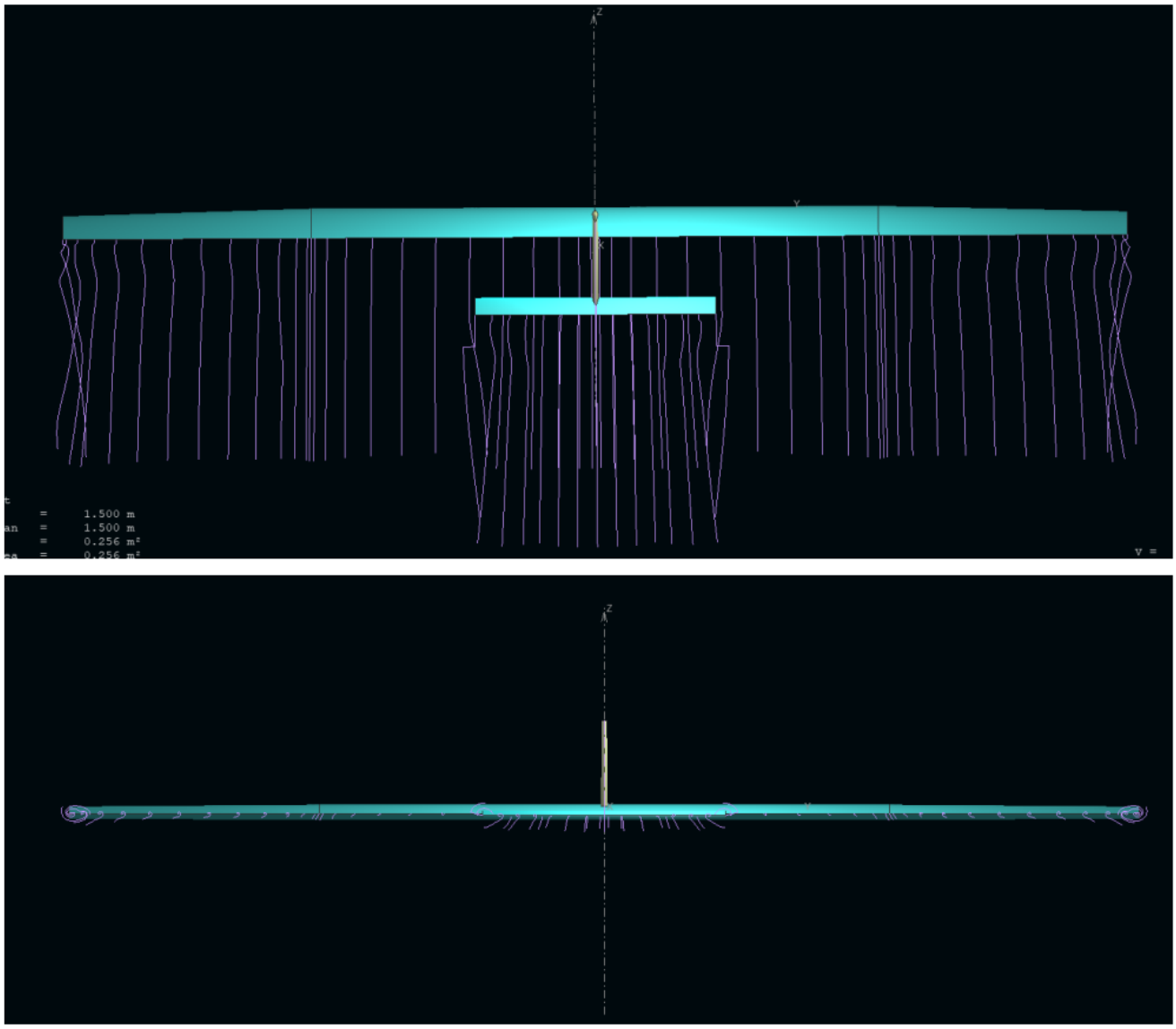


Figure 17: Streamlines over the Wings

Main Aerodynamic Parameters of Wing Analysis[5]

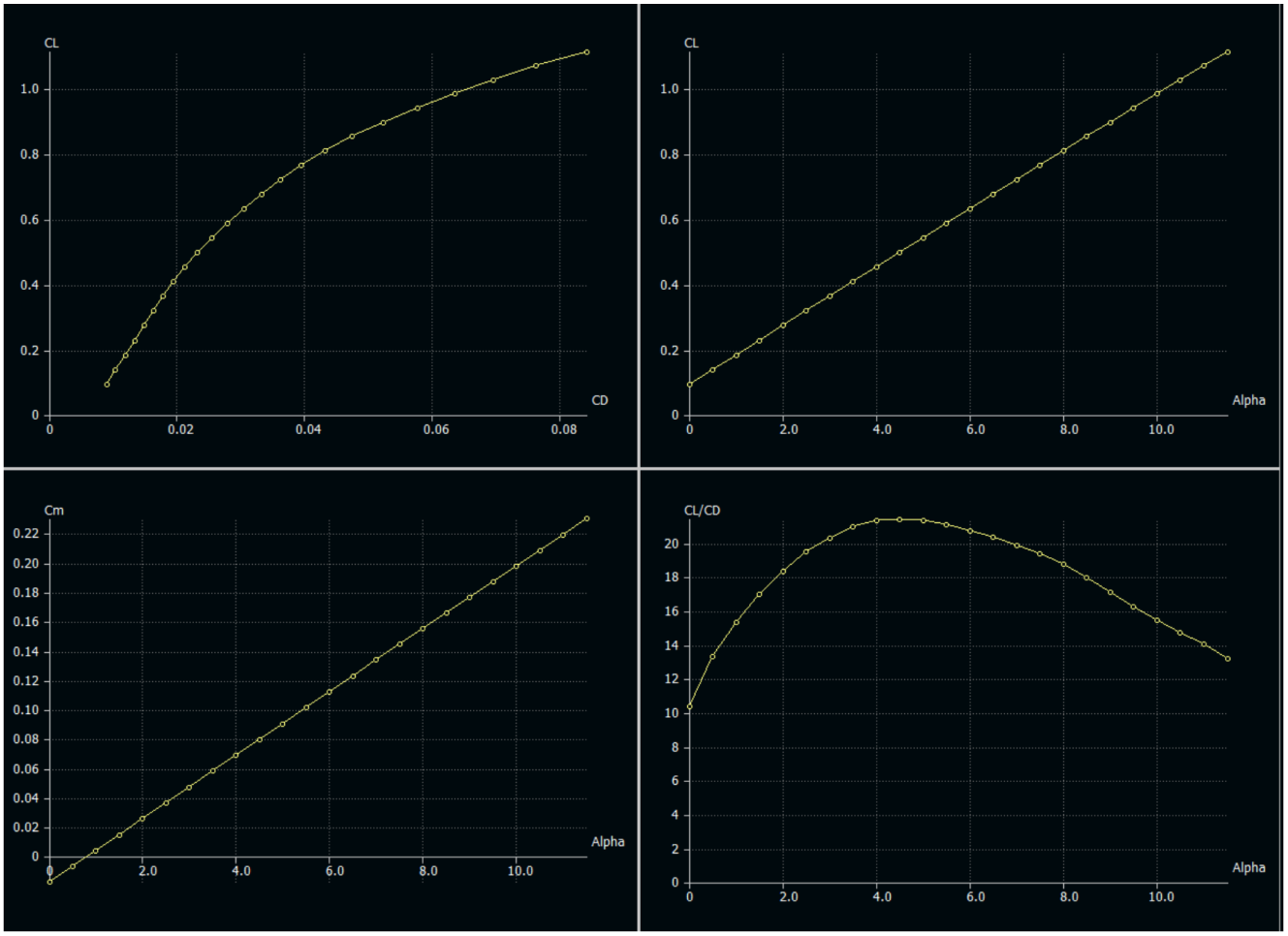


Figure 18: Wing Analysis

Chapter 5: CAD Design in SolidWorks

Fuselage Design[7][8]

After concluding the wing analysis, the subsequent phase involved the design of the fuselage using SolidWorks. The primary goal was to formulate a fuselage configuration adept at accommodating payloads and the complete avionic system. Simultaneously, a detailed Computer-Aided Design (CAD) model of the wings was crafted in SolidWorks, and this component was integrated with the entire aircraft. This integration marked the culmination of the final aircraft design—a cohesive assembly of the fuselage, wings, and associated components.

The design of the fuselage was driven by the imperative to create a versatile structure capable of housing essential elements such as payloads and the avionic system. This component of the aircraft plays a crucial role in determining its overall functionality, considering the accommodation of various equipment and ensuring an optimal weight distribution for stability.

In parallel, the detailed CAD model of the wings, developed in SolidWorks, underwent meticulous scrutiny to ensure precision and accuracy in its representation. This model was then seamlessly incorporated into the broader context of the aircraft assembly. The integration process aimed at achieving a harmonious coordination between the fuselage, wings, and other associated components.

The result was a final aircraft design characterized by a well-coordinated and seamless structure. This comprehensive integration of the fuselage, wings, and associated components laid the groundwork for subsequent stages of development and testing. The SolidWorks environment facilitated the creation of a virtual prototype, offering a realistic representation of the aircraft's form and functionality. This final design served as a critical blueprint for further refinement, testing, and eventual realization of the Micro Aerial Vehicle (MAV), emphasizing the importance of a systematic and integrated approach in the field of aircraft design.

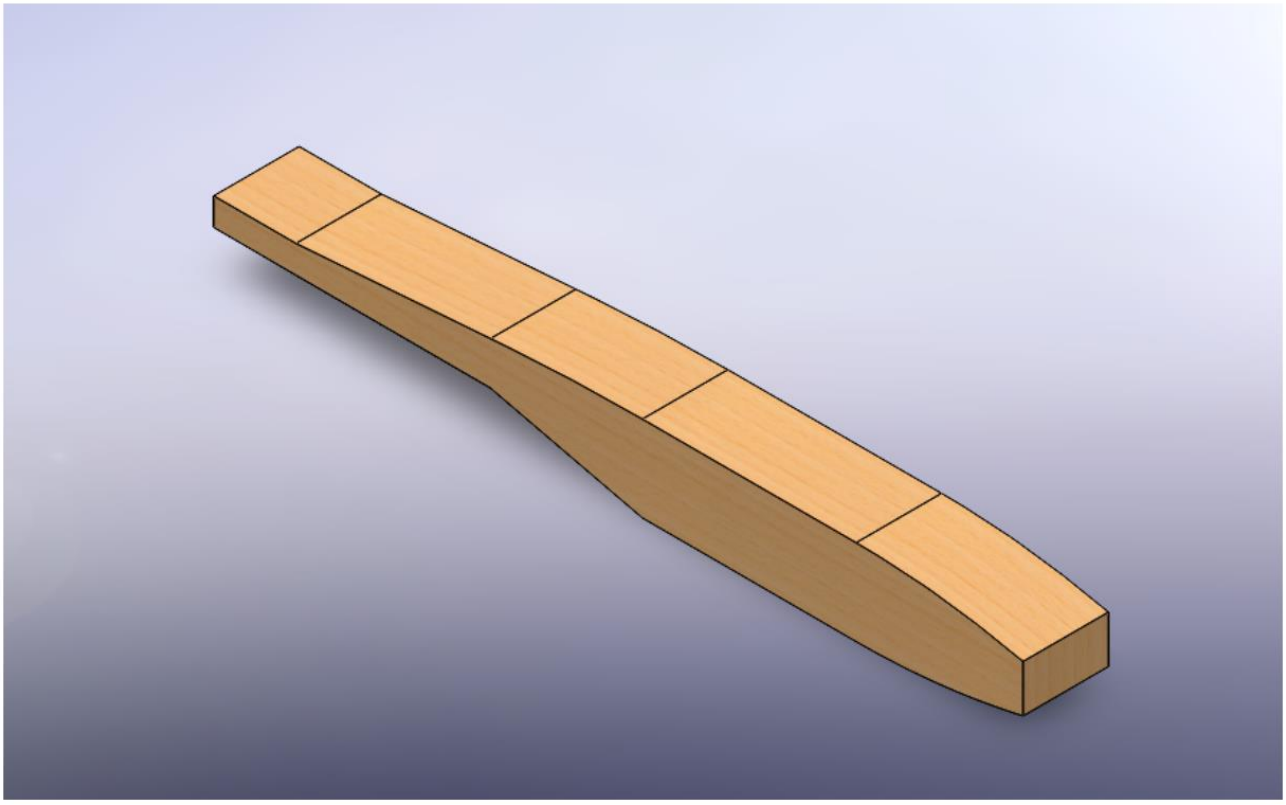


Figure 19: Fuselage

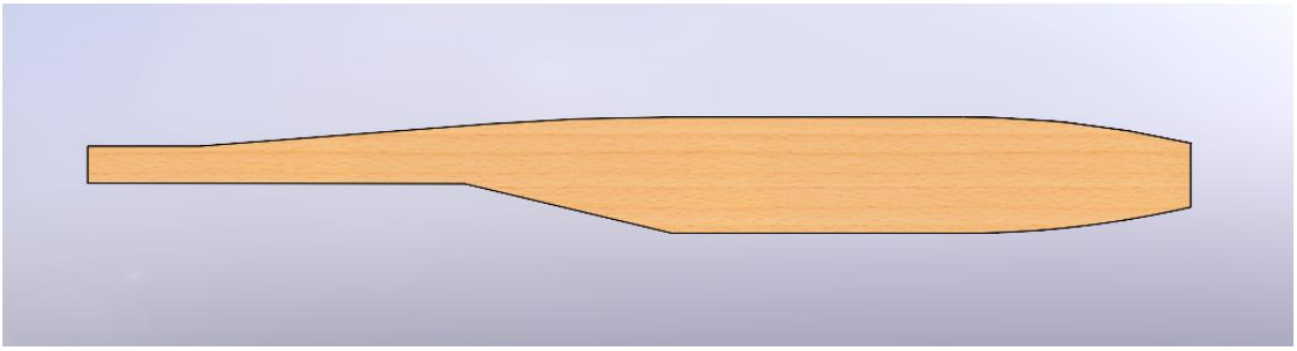


Figure 20: Fuselage Side View

Final Design with Main Wing and Tail Wing[7][8][9]

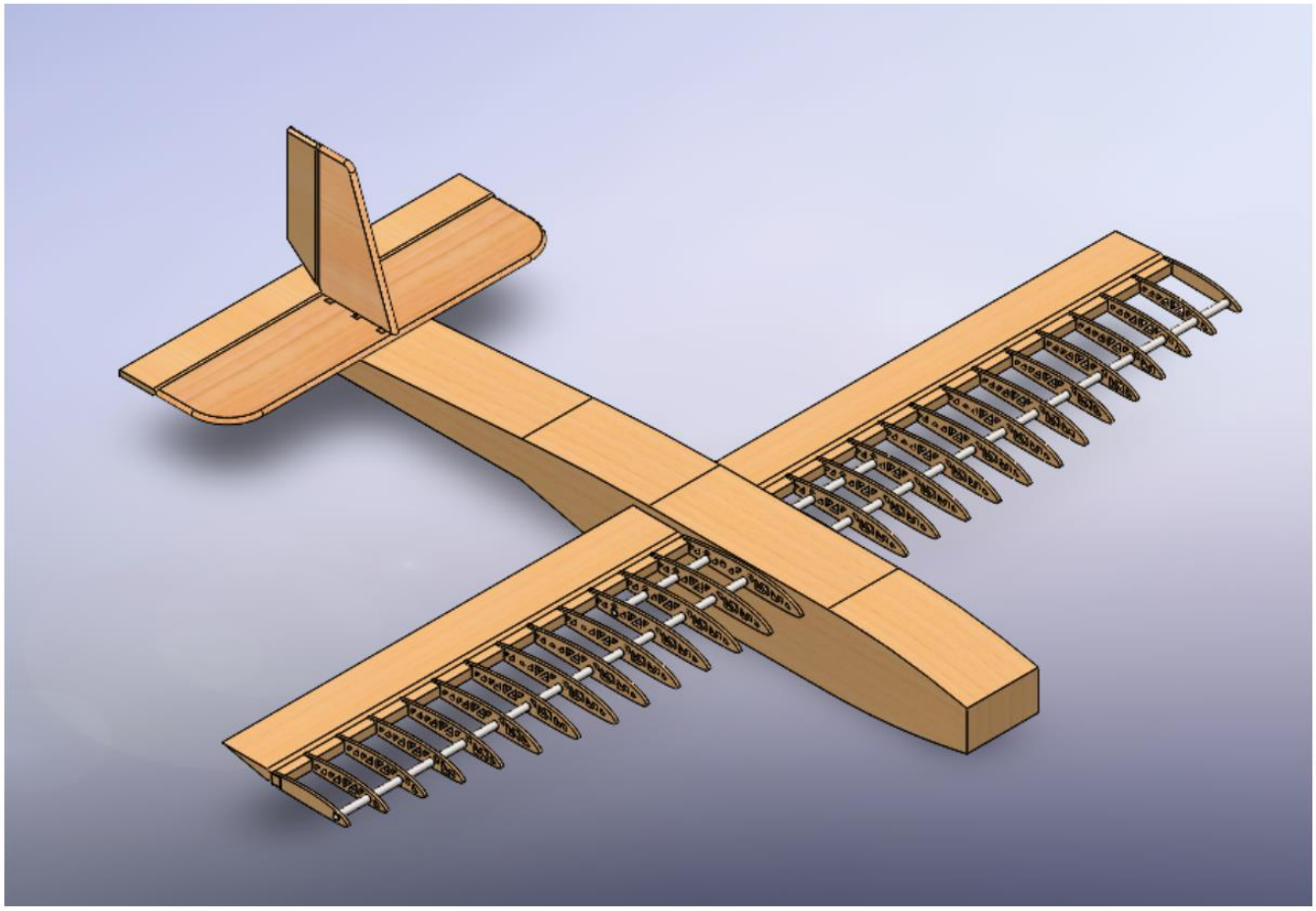


Figure 21: Final Design

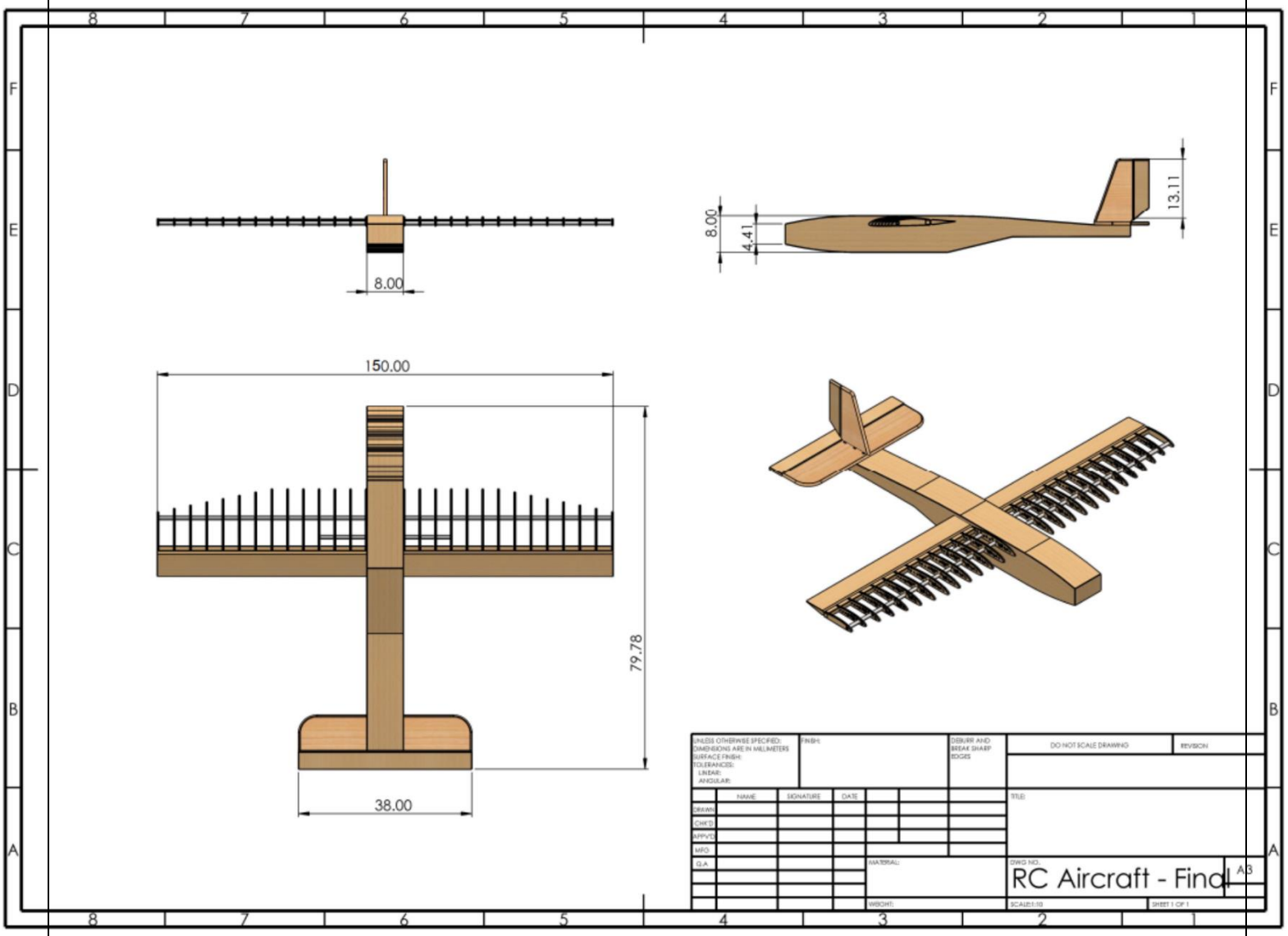


Figure 22: Aircraft Design

CONCLUSION

In conclusion, my internship at Feynman Aerospace has been an invaluable experience in the field of Aircraft Design. Over the span of two months, I immersed myself in various aspects of this discipline, gaining insights into airfoil analysis, wing design, and the complexities involved in crafting a Micro Aerial Vehicle (MAV).

The initial phase of the internship centered around the critical task of airfoil selection and analysis. This involved a meticulous process of optimizing performance while taking into account material cost and weight considerations. The goal was to strike a balance between efficiency and practicality in the design choices made.

The subsequent wing analysis phase brought to light key aerodynamic parameters of the aircraft, including the Coefficient of Lift (CL), Coefficient of Drag (CD), Coefficient of Moment (CM), and CL/CD ratios. This detailed understanding allowed me to refine and enhance the overall design, ensuring the aircraft's performance met the specified criteria.

The CAD design phase, executed in SolidWorks, marked the integration of the fuselage and wings. This crucial step resulted in the creation of a cohesive and ready-to-develop aircraft structure. While the physical construction of the aircraft was not realized during the internship, the project equipped me with a profound understanding of aircraft design principles and honed practical skills essential in this industry.

I express my sincere gratitude to Feynman Aerospace and all individuals who contributed to this enriching learning experience. This internship has undeniably paved the way for my future endeavors in the field of Aircraft Design. I eagerly anticipate applying the knowledge and skills acquired here to contribute meaningfully to real-world aviation projects. This experience has been a stepping stone, solidifying my passion for aircraft design and propelling me toward a promising future in the aviation industry.

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